Johnny Rhe

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Research Interests

I am currently pursuing a Ph.D. degree in the Department of Electrical and Computer Engineering at Sungkyunkwan University. My research focuses on algorithm-hardware co-design for efficient In-Memory Computing (IMC)-based Deep/Convolutional Neural Networks (D/CNNs) inference. I have worked on a wide range of topics including mapping optimization via pruning and compression techniques, reliability enhancement on analog IMC arrays, and a software simulation framework. Recently, I have expanded my research interests to transformer-based models like Vision Transformer (ViT) and Large Language Models (LLMs), aiming to optimize their performance and inference efficiency in IMC systems. To achieve this, I am exploring various hardware-aware optimization techniques, with the ultimate goal of enabling scalable and energy-efficient deployment of these models on IMC systems.

Education

Ph.D. Electrical and Computer Engineering — *Sept. 2019 – Present*

Sungkyunkwan University (SKKU), Suwon, Korea

- GPA: 3.8/4.5
- Intelligent & Resource-Efficient Image Processing & Systems Design (IRIS) Lab
- **Thesis Advisor: Prof. Jong Hwan Ko**

B.S. School of Electronic and Electrical Engineering — *Mar. 2011 - Feb. 2019*

SungKyunKwan University (SKKU), Suwon, Korea

- GPA: 3.8/4.5 (Top 22%)

Research Experience

Accelerated Computation Algorithms for IMC-based D/CNN Inference — *Sept. 2019 - Present*

- Developed novel DNN weight mapping methods [**C2**, **C15**]
- Proposed multi-level mapping optimization via pruning [**J2**, **C4**, **J5**, **C5**, **J3**]
- Proposed low-rank and kernel grouping methods for compact weight representation [**C11**, **C7**]
- Proposed input-stationary mapping methods for energy-efficent and flxible inference [**C3**, **J4**]

Reliability Enhancement on Analog IMC Arrays — *Sept. 2022 - 2024*

- Designed fault-robust remapping schemes to mitigate inference error [**C5**]
- Proposed hybrid grouping to improve multi-bit weight precision and fault tolerance [**C12**]

Software-based Simulator for IMC-based Inference — *Sept. 2021 - 2023*

- Designed visualization and evaluation frameworks integrated with NeuroSim [**C9**]

Resource-efficient Neural Networks for Edge Applications — *Sept. 2023 - 2024*

- Proposed lightweight CNN/SNN models for biomedical signal decoding [**C13**, **C8**]

Projects

A Study on Collaborative Mapping and Training Techniques for Energy-Efficient In-Memory CNN Inference — *Sept. 2024 - Present*

National Research Foundation (NRF), Korea

- **[G1]** Project Leader

Optimizing DNN Mapping and Execution on In-Memory Computing Arrays — *April. 2024 - Present*

Ministry of Trade, Industry & Energy (MOTIE), Korea

HW-SW Co-Design for Low-Power Real-Time In-Memory Deep Learning Operation — *Mar. 2023 - Present*

Samsung Advanced Institute of Technology (SAIT), Korea

Spatiotemporal Precision Control for Efficient and Robust In-Memory Deep Learning Inference — *June. 2023 - May. 2024*

National Research Foundation (NRF), Korea

A Co-Optimization Simulation Framework for Efficient In-Memory Deep Learning Computation — *June. 2022 - May. 2023*
National Research Foundation (NRF), Korea

Energy-Efficient DNN Design on Memristor-Based IC platforms — *Sept. 2020 - Mar. 2023*
Institute for Information & communications Technology Planning & Evaluation (IITP), Korea

Honors & Awards

[H16] $2^{nd}/3^{rd}$ Place Winners at Grand Challenge on Brain-Machine Interfaces — *Oct, 2025*
IEEE Biomedical Circuits and Systems Conference (BioCAS)

[H15] 2025 SKKU Innovative Research Fellowship — *May, 2025*
Sungkyunkwan University (SKKU)

[H14] Poster Presentation at PhD Forum — *Mar, 2025*
Design, Automation and Test in Europe Conference (DATE) 2025

[H13] Outstanding Research Award — *Nov, 2024*
Korea Collaborative & High-tech Initiative for Prospective Semiconductor Research (K-CHIPS)

[H12] KETEP Outstanding Research Award — *Oct, 2024*
Korea Institute of Energy Technology Evaluation and Planning (KETEP)

[H11] Samsung Outstanding Paper Award - Samsung Elec. & SKKU (2024) — *Aug, 2024*
Samsung Electronics

[H10] Grand Prize at the Exynos AI Challenger — *July, 2024*
Samsung Electronics' S.LSI

[H9] 3^{rd} Place Winner in the System Design Contest (GPU Track) — *June, 2024*
Design Automation Conference (DAC) 2024

[H8] Poster Presentation at PhD Forum — *May, 2024*
IEEE International Symposium on Circuits and Systems (ISCAS) 2024

[H7] Most Popular Poster Award at the Student Research Forum — *Jan, 2024*
29th Asia South Pacific Design Automation Conference (ASP-DAC) 2024

[H6] Poster Presentation at Student Research Forum — *Jan, 2024*
29th Asia South Pacific Design Automation Conference (ASP-DAC) 2024

[H5] Samsung Best Paper Award - Samsung Elec. & SKKU (2023) — *Aug, 2023*
Samsung Electronics

[H4] IEEE CASS Non-Flagship Conference Student Travel Grant (Selection) - ISLPED — *Aug, 2023*
IEEE Circuits and Systems Society (CAS)

[H3] Best Paper Award — *July, 2023*
Korean Artificial Intelligence Association (KAIA)

[H2] 2022 SKKU Innovative Research Fellowship — *Sept, 2022*
Sungkyunkwan University (SKKU)

[H1] 2^{nd} Place Winner of Artificial Intelligence Grand Challenge — *July, 2020*
Institute of Information & Communications Technology Planning & Evaluation, Korea

Featured Publications

Journal Articles

[J5] **Johnny Rhe**, Kang Eun Jeon and Jong Hwan Ko, "Genetic Algorithm-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping," *Journal of Systems Architecture (JSA)*, May. 2025 (JCR Q1). [Paper]

[J3] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seong Moon Jeong, and Jong Hwan Ko, "KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping," *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I) vol. 71, no. 12, pp. 6138–6151*, Nov. 2024 (JCR Q1). [Paper]
[H12]Outstanding Research Award from KETEP (2024)

[J2] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, "VWC-SDK: Convolutional Weight Mapping Using Shifted and Duplicated Kernel with Variable Windows and Channels," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS) vol. 12, no. 2, pp. 408–421*, May. 2022 (JCR Q1). [Paper]

Conference Proceedings

[C15] **Johnny Rhe**, Juhong Park, Kang Eun Jeon, and Jong Hwan Ko, "ETA: Efficient Transformer Attention Mapping for ReRAM-based Compute-In-Memory Architectures," *Asia Pacific Conference on Circuits and Systems (APCCAS) (Accepted)*, Oct. 2025.

[C5] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seong Moon Jeong, and Jong Hwan Ko, "Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays," *ACM/IEEE International Conference on Computer-Aided Design (ICCAD)*, Oct. 2023. [Paper]
[H11] Outstanding Paper Award from Samsung Elec.(2024)

[C4] **Johnny Rhe**, Kang Eun Jeon, and Jong Hwan Ko, "PAIRS: Pruning-AIded Row-Skipping for SDK-Based Convolutional Weight Mapping in Processing-In-Memory Architectures," *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, Aug. 2023. [Paper]
[H4] Student Travel Grant Selection from IEEE CAS

[C2] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, "VW-SDK: Efficient Convolutional Weight Mapping Using Variable Windows for Processing-In-Memory Architectures," *Design, Automation, and Test in Europe (DATE)*, Mar. 2022 (Oral, acceptance rate: 25%). [Paper]

Full Publications (110+ Citations)

Most-up-to-date publication list can be founded at Google Scholar

Journal Articles

[J5] **Johnny Rhe**, Kang Eun Jeon and Jong Hwan Ko, "Genetic Algorithm-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping," *Journal of Systems Architecture (JSA)*, May. 2025 (JCR Q1). [Paper]

[J4] Juhong Park, **Johnny Rhe**, and Jong Hwan Ko, "Input/Mapping Precision Controllable Digital PIM with Adaptive Adder Tree Architecture for Flexible DNN Inference," *Journal of Systems Architecture (JSA)*, Feb. 2025 (JCR Q1). [Paper]

[J3] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seong Moon Jeong, and Jong Hwan Ko, "KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping," *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I) vol. 71, no. 12, pp. 6138–6151*, Nov. 2024 (JCR Q1). [Paper]
[H12] Outstanding Research Award from KETEP (2024)

[J2] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, "VWC-SDK: Convolutional Weight Mapping Using Shifted and Duplicated Kernel with Variable Windows and Channels," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS) vol. 12, no. 2, pp. 408–421*, May. 2022 (JCR Q1). [Paper]

[J1] Eunyoung Lee, Taeyoung Han, Donguk Seo, Gicheol Shin, Jaerok Kim, Seonho Kim, Soyoun Jeong, **Johnny Rhe**, Jaeyung Park, Jong Hwan Ko, Yoonmyung Lee, "A Charge-Domain Scalable-Weight In-Memory Computing Macro with Dual-SRAM Architecture for Precision-Scalable DNN Accelerators," *IEEE Transactions on Circuits and Systems I (TCAS-I) vol. 68, no. 8, pp. 3305–3316*, May. 2021 (JCR Q1). [Paper]

Conference Proceedings

[C15] **Johnny Rhe**, Juhong Park, Kang Eun Jeon, and Jong Hwan Ko, "ETA: Efficient Transformer Attention Mapping for ReRAM-based Compute-In-Memory Architectures," *Asia Pacific Conference on Circuits and Systems (APCCAS) (Accepted)*, Oct. 2025.

[C14] **Johnny Rhe**, and Jong Hwan Ko, "Weight Sharing for Array-Efficient CNN Inference in Compute-In-Memory Architectures," *International SoC Design Conference (ISOCC) (Accepted)*, Oct. 2025.

[C13] Chanwook Hwang, Chan Woong Park, **Johnny Rhe**, and Jong Hwan Ko, "Efficient and Robust SNN Decoder Training for Closed-Loop Brain-Machine Interfaces," *IEEE Biomedical Circuits and Systems Conference (BioCAS) (Accepted)*, Oct. 2025.
[H16] 2nd/3rd Place Winner at Grand Challenge by BioCAS

[C12] Kang Eun Jeon, Sangheum Yeon, Jinhee Kin, Hyeonsu Bang, **Johnny Rhe**, and Jong Hwan Ko, "Row-Column Hybrid Grouping for Fault-Resilient Multi-Bit Weight Representation on IMC Arrays," *ACM/IEEE International Conference on Computer-Aided Design (ICCAD)*, Oct. 2025. [Paper]

[C11] Kang Eun Jeon, **Johnny Rhe**, and Jong Hwan Ko, "Low-Rank Compression for IMC Arrays," *Design, Automation & Test in Europe Conference (DATE)*, Mar. 2025. [Paper]

[C10] **Johnny Rhe**, and Jong Hwan Ko, "Row-Efficient Pruning for In-Memory Convolutional Weight Mapping," *International SoC Design Conference (ISOCC)*, Aug. 2024. [Paper]

[C9] Kang Eun Jeon[1], Wooram Seo[1], **Johnny Rhe**, and Jong Hwan Ko, "ConvMapSim: Modeling and Simulating Convolutional Network Mapping on PIM Arrays," *IEEE Artificial Intelligence Circuits and Systems (AICAS)*, Apr. 2024. [Paper]

[C8] Chanwook Hwang, Jaehyeon So, **Johnny Rhe**, Jiyoon Kim, Juhong Park, Kang Eun Jeon, and Jong Hwan Ko, "An Efficient Ventricular Arrhythmias Detection on Microcontrollers with Optimized 1D CNN," *IEEE Artificial Intelligence Circuits and Systems (AICAS)*, Apr. 2024. [Paper]
[H10] Grand Prize at Exynos AI Challenger by Samsung Elec.

[C7] Juhong Park, **Johnny Rhe**, and Jong Hwan Ko, "KARS: Kernel-Grouping Aided Row-Skipping for SDK-based Weight Compression in PIM Arrays," *IEEE International Symposium on Circuits and Systems (ISCAS)*, May 2024. [Paper]
[H13] Best Paper Award from KAIA (2023)

[C6] Hyeonsu Bang, Kang Eun Jeon, **Johnny Rhe**, and Jong Hwan Ko, "DCR: Decomposition-Aware Column Re-Mapping for Stuck-At-Fault Tolerance in ReRAM Arrays," *IEEE International Conference on Computer Design (ICCD)*, Nov. 2023. [Paper]

[C5] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seong Moon Jeong, and Jong Hwan Ko, "Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays," *ACM/IEEE International Conference on Computer-Aided Design (ICCAD)*, Oct. 2023. [Paper]
[H11] Outstanding Paper Award from Samsung Elec. (2024)

[C4] **Johnny Rhe**, Kang Eun Jeon, and Jong Hwan Ko, "PAIRS: Pruning-AIded Row-Skipping for SDK-Based Convolutional Weight Mapping in Processing-In-Memory Architectures," *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, Aug. 2023. [Paper]
[H4] Student Travel Grant Selection from IEEE CAS

[C3] Kang Eun Jeon, **Johnny Rhe**, and Jong Hwan Ko, "Weight-Aware Activation Mapping for Energy-Efficient Convolution on PIM Arrays," *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, Aug. 2023. [Paper]
[H5] Outstanding Paper Award from Samsung Elec. (2023)

[C2] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, “VW-SDK: Efficient Convolutional Weight Mapping Using Variable Windows for Processing-In-Memory Architectures,” *Design, Automation, and Test in Europe (DATE)*, Mar. 2022 (Oral, acceptance rate: 25%). [Paper]

[C1] Gicheol Shin, Donguk Seo, Jaerok Kim, **Johnny Rhe**, Eunyoung Lee, Seonho Kim, Soyoun Jeong, Jong Hwan Ko, Yoonmyung Lee, “A charge-Domain Computation-in-Memory Macro with Versatile All-Around-Wire-Capacitor for Variable-Precision Computation and Array-Embedded DA/AD Conversions,” *European Conference on Solid-State Circuits (ESSCIRC)*, Sept. 2021. [Paper]

Patents

[P9] **Johnny Rhe**, Kang Eun Jeon, and Jong Hwan Ko, US Patent Application No.19069410, method and electronic device with weight pruning cross-reference to related application, Mar. 2025

[P8] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, and Jong Hwan Ko, PCT Patent Application No.PCT/KR2024/016234, Method and apparatus for convolution operation utilizing kernel shape control, Oct. 2024

[P7] **Johnny Rhe**, Kang Eun Jeon, and Jong Hwan Ko, KR Patent Application No.10-2024-0058360, Pattern-based weight pruning method and electronic devices, May. 2024

[P6] **Johnny Rhe**, and Jong Hwan Ko, US Patent Application No.18398389, Apparatus and method for controlling process-in-memory by accelerating convolution operation based on arranging pattern of weight in kernel, and storage medium storing instructions to perform method for controlling process-in-memory, Dec. 2023

[P5] Juhong Park, **Johnny Rhe**, and Jong Hwan Ko, KR Patent Application No.10-2023-0157705, Adaptive adder tree and digital in-memory architecture supporting multi-bit partitioning and improved column-wise mapping for depth-wise convolution layers, Nov. 2023

[P4] **Johnny Rhe**, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, and Jong Hwan Ko, KR Patent Application No.10-2023-0171519, Method and apparatus for convolution operation utilizing kernel shape control, Nov. 2023

[P3] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, US Patent Application No.18090628, Memory device for optimizing computation of convolution layer, method for controlling memory device, and recording medium storing instruction to perform method for controlling memory device, Dec. 2022

[P2] **Johnny Rhe**, and Jong Hwan Ko, KR Patent Application No.10-2022-0187676, Apparatus and method for controlling process-in-memory by accelerating convolution operation based on arranging pattern of weight in kernel, and storage medium storing instructions to perform method for controlling process-in-memory, Dec. 2022

[P1] **Johnny Rhe**, Sungmin Moon, and Jong Hwan Ko, KR Patent No.No.10-2021-0191289, Memory device for optimizing computation of convolution layer, method for controlling memory device, and recording medium storing instruction to perform method for controlling memory device, Dec. 2021

Talks

Optimizing Weight Mapping for Energy-Efficient In-Memory CNN Inference — *Oct, 2025*
International Conference on Computer-Aided Design (ICCAD)

ETA: Efficient Transformer Attention Mapping for ReRAM-based Compute-In-Memory Architectures — *Oct, 2025*
Asia Pacific Conference on Circuits and Systems (APCCAS)

Enabling Resource-Efficient AI via HW-SW Co-Design — *Aug, 2025*
University of Illinois Chicago (UIC)

Enabling Resource-Efficient AI via HW-SW Co-Design — *Aug, 2025*
Duke University

Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays — *Oct, 2023*
International Conference on Computer Aided Design (ICCAD)

Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays — *Sept, 2023*
Sungkyunkwan University (SKKU) AI Colloquium

PAIRS: Pruning-Aided Row-Skipping for SDK-Based Convolutional Weight Mapping in Processing-In-Memory Architectures — *Aug, 2023*
International Symposium on Low Power Electronics and Design (ISLPED)

VWC-SDK: Convolutional Weight Mapping Using Shifted and Duplicated Kernel with Variable Windows and Channels — *Sept, 2022*
Sungkyunkwan University (SKKU) AI Colloquium

VW-SDK: Efficient Convolutional Weight Mapping Using Variable Windows for Processing-In-Memory Architectures — *Mar, 2022*
Design, Automation, and Test in Europe (DATE)

Teaching Experience

Teaching Assistant in Operating System — *Mar, 2020*
Sungkyunkwan University (SKKU), Suwon, Korea
- Instructor: Prof. Jong Hwan Ko

Student Guidance

Mentor for Undergraduate Research Project — *Sept. 2025 - Present*
Sungkyunkwan University (SKKU), Suwon, Korea
- Dynamic Computation Adjustment Techniques for IMC-based Efficient CNN Inference
- *Yehyeon Kim, Seongjo Lee*

Graduate Research Mentorship — *Mar. 2023 - Mar. 2024*
Sungkyunkwan University (SKKU), Suwon, Korea
- Kernel-Grouping Techniques for IMC Arrays [C7] - 🏆 **[H12] Best Paper Award from KAIA (2023)**
- *Juhong Park*

Mentor for Outstanding Undergraduate Research Program — *Feb. 2022 - June. 2022*
Sungkyunkwan University (SKKU), Suwon, Korea
- Remapping Method for Mitigating Cell Variation in IMC Arrays
- *Minho Lee, Minseop Kim, Seonghoon Jeong, Jihoon Choi*

Grants

Development of Collaborative Intelligent Agent Technologies for Human–AI Symbiotic Cooperation — *July, 2025*
Sungkyunkwan University (SKKU), Suwon, Korea
- Contributed to concept development and writing of the proposal
- Institute for Information & communications Technology Planning & Evaluation (IITP), Korea (Project # RS-2025-25442569)
- Amount awarded: ₩ 11.0 billion ($ 8.15 million)

A Study on Collaborative Mapping and Training Techniques for Energy-Efficient In-Memory CNN Inference — *July, 2024*
Sungkyunkwan University (SKKU), Suwon, Korea
- Core idea development and proposal writing as a project leader
- National Research Foundation (NRF), Korea (Project # RS-2024-00409161)
- Amount awarded: ₩ 25 million ($ 17.9 thousand)

Technical Reviewer Activities

IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)

Design Automation Conference (DAC)

Skills

Programing Python, PyTorch, VHDL, Verilog
Tools NeuroSim

References

Dr. Jong Hwan Ko (*Ph.D. advisor*)

Associate Professor, *Dept. of Electrical and Computer Engineering*,
Sungkyunkwan University (SKKU), Suwon, Korea
https://iris.skku.edu/ jhko@skku.edu

Dr. Kang Eun Jeon

Postdoctoral Researcher, *Kim Jaechul Graduate School of Artificial Intelligence*,
Korea Adavanced Institute of Science & Technology (KAIST), Seongnam, Korea
Google Scholar kejeon@kaist.ac.kr